

1. Simple Linear Search Program

Step-by-step explanation:

PROGRAM:

```
#include <iostream>

using namespace std;

int main()
{
    int arr[5] = {10, 20, 30, 40, 50};

    int key;

    int found = 0;

    cout << "Enter number to search: ";

    cin >> key;

    // Check each element one by one
    for(int i = 0; i < 5; i++)
    {
        if(arr[i] == key)
        {
            cout << "Element found at position " << i + 1;

            found = 1;

            break;
        }
    }

    if(found == 0)
    {
        cout << "Element not found";
    }

    return 0;
}
```

Output of Linear Search

Enter number to search: 30

Element found at position 3

If the element is not present:

Enter number to search: 25

Element not found

2. Simple Binary Search Program

PROGRAM:

```
#include <iostream>
using namespace std;

int main()
{
    int arr[5] = {10, 20, 30, 40, 50};

    int key;
    int low = 0;
    int high = 4;
    int found = 0;

    cout << "Enter number to search: ";
    cin >> key;

    while(low <= high)
    {
        int mid = (low + high) / 2;

        if(arr[mid] == key)
        {
            cout << "Element found at position " << mid + 1;
```

```
        found = 1;
        break;
    }
    else if(key > arr[mid])
    {
        low = mid + 1;
    }
    else
    {
        high = mid - 1;
    }
}

if(found == 0)
{
    cout << "Element not found";
}

return 0;
}
```

OUTPUT:

Output of Binary Search

Enter number to search: 40

Element found at position 4

If the element is not present:

Enter number to search: 25

Element not found